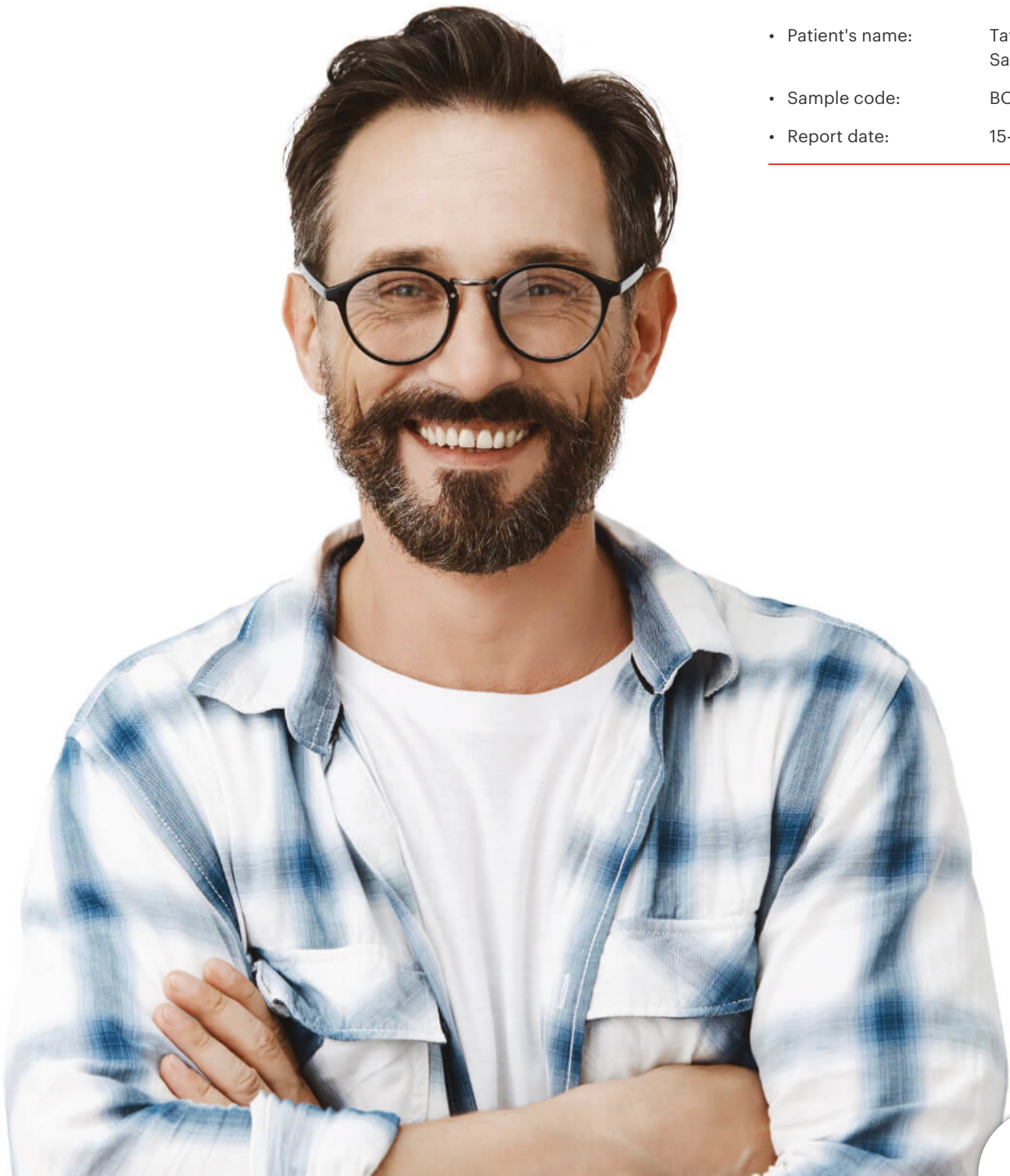




• Patient's name:	Tafhim Mohammed Salman
• Sample code:	BOX01036AATRI
• Report date:	15-11-2023



Fagron TrichoTest™

Results report



TrichoTest™ Genetic report

LEGAL DISCLAIMER

Fagron Genomics, S.L.U carries out genetic tests upon request by healthcare professionals, in relation to biological samples from patients obtained by the healthcare professional. Our tests do not replace a medical consultation, nor do they make up a diagnostic or treatment, nor should they be interpreted this way. Only healthcare professionals can interpret the results of said tests, based on their knowledge of the clinical records of the patients and other relevant factors and, under their responsibility, give a diagnostic or prescribe treatment to the patient. We decline all responsibility derived from the use and interpretation of the results of our tests by the solicitant healthcare professional. Fagron Genomics, S.L.U expressly reserves any legal actions in case of an innapropriate, negligent or incorrect use or interpretation of the results of our tests. It is the responsibility of the healthcare professional who requests a test to guarantee to the patient the appropriate genetic advice as foreseen by Law 14/2007, of 3rd July, of biomedical research. As Fagron Genomics, S.L.U does not have access to the personal identifiable information about the patient from whom the sample comes, it is the responsibility of the requesting healthcare professional to comply with the applicable data protection Laws and regulations.



I. Patient identification data

1.

Patient identification data



Ordering physician —●— **Dr. Sezgin Cagatay**

Contact —●— **yasmina.mendhao@hortmanclinics.com**

Patient's name —●— **Tafhim Mohammed Salman**

Gender —●— **Male**

Date of birth —●— **21-03-1996**

Sample type —●— **Buccal Swab**

Sample code —●— **BOX01036AATRI**

Sample date —●— **08-11-2023**

Report date —●— **15-11-2023**



II. Recommendation of the most suitable drugs and supplements

2.

Recommendation of the most suitable drugs and supplements

The **genetic test** uses an automated qualitative pharmacogenetic algorithm that analyzes the patient's genetic data and combines this information with relevant patient history to recommend the most suitable active ingredients. Next, we show on a color scale which compounds the algorithm recommends the most. The transition from white to dark green indicates drugs from least recommended to most recommended. Medications blocked due to intolerances or contraindications are shown in red.

Anti-alopecic drugs

Prostaglandins	
• Latanoprost	67%
• Oral Minoxidil (man)	65%
• Minoxidil	64%
• Prostaquinon TM	50%
• Cetirizine Hcl	50%
• Oral Minoxidil (woman)	

Antiandrogenic	
• Dutasteride	100%
• Finasteride	100%
• Saw Palmetto	100%
• Oral Finasteride (man)	100%
• Oral Dutasteride	97%
• Melatonin	75%
• 17- α Estradiol	
• Spironolactone	
• Oral Finasteride (woman)	

Anti-inflammatory	
• Triamcinolone acetonide	
• Hydrocortisone	
• Betamethasone dipropionate	
• Fluocinolone acetonide	

Immunomodulator	
• Tacrolimus	

Hair care supplements

Circulation	
• Arginine	85%
• L-Carnitine L-tartrate	66%

Collagen synthesis	
• Oral SiliciuMax TM	
• Cystine	

Insulin-like growth factor increase	
• IGrantine-F1 TM	

Blocked



Recommended



• Patient name: **Tafhim Mohammed Salman**

• Patient ID: **BY0710109**

• Date of Birth: **21-03-1996**

• Sample code: **BOX01036AATRI**

• Sample date: **08-11-2023**

• Date of the results: **13-11-2023**

Vitamin, mineral and antioxidant supplements

Vitamin deficiency	
• Vitamin B12 (Cianocobalamin)	100%
• Vitamin E (Tocoferol)	100%
• Vitamin B7 (Biotin)	
• Vitamin B7 (Biotin)	
• Retinol palmitate	
• Vitamin D	
• Vitamin B9 (Folate)	

Softener	
• D-Panthenol	70%

Minerals	
• Zinc acetate	100%

Keratolytic	
• Tretinoin	

Blocked



Recommended





III. Formulas for personalized treatment

• Patient name: **Tafhim Mohammed Salman**

• Patient ID: **BY0710109**

• Date of Birth: **21-03-1996**

• Sample code: **BOX01036AATRI**

• Sample date: **08-11-2023**

• Date of the results: **13-11-2023**

3. Formulas for personalized treatments

The pharmacogenetic algorithm has selected a series of formulations for topical, oral use or capillary mesotherapy for the care and hygiene of your patient's scalp. These personalized formulations have been selected taking into account the genetics, the type of alopecia, and the relevant history of the patient.

Oral treatment

Formula

Oral Minoxidil (man)	3.25 mg
Oral Finasteride (man)	1 mg
Vitamin B12 (Cianocobalamin)	1000 mcg

Posology

1 capsule per day, 90 capsules for 3 months

Signature of the prescribing physician

Dr

Physician registration No.

Date

Address

Signature

• Patient name: **Tafhim Mohammed Salman**

• Patient ID: **BY0710109**

• Date of Birth: **21-03-1996**

• Sample code: **BOX01036AATRI**

• Sample date: **08-11-2023**

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Topical treatment

Formula

Prostaquinon TM 2 %

Dutasteride 0.25 %

Arginine 2 %

TrichoSol 100ml

Posology

Apply at night before bedtime. Leave the solution on your scalp for as long as possible. Wash your scalp the next day.

Signature of the prescribing physician

Dr

Physician registration No.

Date

Address

Signature

• Patient name: **Tafhim Mohammed Salman**

• Patient ID: **BY0710109**

• Date of Birth: **21-03-1996**

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Scalp care and hygiene

Topical treatment

Formula

Zinc acetate 0.2 %

Topical Saw Palmetto 2 %

Vitamin E (Tocoferol) 5 %

TrichoOil 30ml

Posology

1-2 times / week, massage for 3-5 minutes and leave it on for 10 min before washing your hair.

Signature of the prescribing physician

Dr

Physician registration No.

Date

Address

Signature

• Patient name: **Tafhim Mohammed Salman**

• Patient ID: **BY0710109**

• Date of Birth: **21-03-1996**

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Scalp care and hygiene

Topical treatment

Formula

Zinc acetate 0.2 %

Arginine 2 %

D-Panthenol 0.43 %

TrichoWash 250ml

Posology

Massage for 2 minutes and rinse

Signature of the prescribing physician

Dr

Physician registration No.

Date

Address

Signature

• Patient name: **Tafhim Mohammed Salman**

• Patient ID: **BY0710109**

• Date of Birth: **21-03-1996**

• Sample code: **BOX01036AATRI**

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Scalp care and hygiene

Topical treatment

Formula

Arginine

2 %

TrichoCond

250ml

Posology

After washing your hair, apply the conditioner and leave it on for 2-3 minutes before rinse.

Signature of the prescribing physician

Dr

Physician registration No.

Date

Address

Signature

• Patient name: **Tafhim Mohammed Salman**

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Scalp care and hygiene

Topical treatment

Formula

Arginine

2 %

TrichoSerum

50ml

Posology

After washing your hair, apply on wet hair.

Signature of the prescribing physician

Dr

Physician registration No.

Date

Address

Signature



IV. Complete data

• Patient name: **Tafhim Mohammed Salman**

• Patient ID: **BY0710109**

• Date of Birth: **21-03-1996**

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• Sample date: **08-11-2023**

• Date of the results: **13-11-2023**

4.

Complete data

Data from the medical questionnaire

Patient demographics

Gender —●— **Male**

Age (years) —●— **27**

Height (cm) —●— **165**

Weight (kg) —●— **85**

BMI —●— **31.22**

Family history of alopecia —●— **Parents**

Hair loss data

Type of alopecia —●— **Androgenic alopecia**

Grade of alopecia —●— **Grade II**

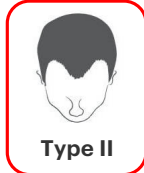
Time elapsed since start of hair loss —●— **+1**

Prescription of testosterone derivatives —●— **No**

Norwood-Hamilton Scale



Type I



Type II



Type III



Vertex



Type IV



Type V



Type VI



Type VII

Clinical examination

Amount of hair loss —●— **Little bit**

Complaints associated with alopecia —●— **Dandruff**

Patchy alopecia —●— **No**

Current anti-alopecia treatment —●— **No**

Previous anti-alopecia treatment —●— **No**

4.

Complete data
Pharmacogenetic results

1. Anti-alopecic drugs

Treatment efficacy with prostaglandin inhibitors

Prostaglandin D2				
Gene	SNP (transition)	Activating allele	Patient genotype	Pharmacogenetic result
GPR44-1	rs545659 (A>G)	G	GG	Genetic result: Predisposition to higher GPR44 mRNA stability. Interpretation: Prostaglandin D2 receptor 2 (GPR44 or CRTH2) variants are associated with an increased GPR44 mRNA stability leading to higher responsiveness to prostaglandin D2 and hair follicle regression. Treatment/dosage: Treatment with prostaglandin D2 inhibitors (Cetirizine and/or Prostaquinon) at high doses would be highly recommended.
GPR44-2	rs533116 (G>A)	A	GG	Genetic result: Predisposition to normal GPR44 mRNA stability. Interpretation: Prostaglandin D2 receptor 2 (GPR44 or CRTH2) variants are associated with an increased GPR44 mRNA stability leading to higher responsiveness to prostaglandin D2 and hair follicle regression. Treatment/dosage: SNP analysis does not indicate the necessity to treat with prostaglandin D2 inhibitors.

Latanoprost				
Gene	SNP (transition)	Activating allele	Patient genotype	Pharmacogenetic result
PTGFR-1	rs6686438 (T>G)	G	GT	Genetic result: Increased likelihood of having a positive response to Latanoprost. Interpretation: Prostaglandin F receptor (PTGFR) variants are related with Latanoprost treatment efficacy (prostaglandin analog) . Treatment/dosage: Treatment with Latanoprost at normal doses is recommended.
PTGFR-2	rs1328441 (G>A)	A	GA	Genetic result: Increased likelihood of having a positive response to Latanoprost. Interpretation: Prostaglandin F receptor (PTGFR) variants are related with Latanoprost treatment efficacy (prostaglandin analog) . Treatment/dosage: Treatment with Latanoprost at normal doses is recommended.
PTGFR-3	rs10782665 (T>G)	G	GT	Genetic result: Increased likelihood of having a positive response to Latanoprost. Interpretation: Prostaglandin F receptor (PTGFR) variants are related with Latanoprost treatment efficacy (prostaglandin analog) . Treatment/dosage: Treatment with Latanoprost at normal doses is recommended.

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Treatment efficacy with minoxidil

Minoxidil				
Gene	SNP (transition)	Activating allele	Patient genotype	Pharmacogenetic result
PTGES2	rs13283456 (C>T)	T	CT	Genetic result: Predisposition to slightly reduced PGE2 levels. Interpretation: Prostaglandin E synthase 2 (PTGES2) variants are associated with lower prostaglandin E2 production (hair growth promoter). Treatment/dosage: Treatment with normal doses of Minoxidil to stimulate prostaglandin E2 would be recommended.
SULT1A1	rs9282861 (C>T)	T	TC	Genetic result: Predisposition to reduced SULT1A activity. Interpretation: Minoxidil Sulfotransferase Enzyme (SULT1A1) variants predict response to minoxidil treatment. Treatment/dosage: Minoxidil may be recommended but in high doses, with the aim of trying to ensure therapeutic activity.

Treatment efficacy with glucocorticoid anti-inflammatories

Glucocorticoides				
Gene	SNP (transition)	Activating allele	Patient genotype	Pharmacogenetic result
GR-alpha	rs6198 (A>G)	G	GA	Genetic result: Predisposition to moderate resistance to glucocorticoid anti-inflammatory treatments. Interpretation: Glucocorticoid Receptor (GR or NR3C1) variants are associated with resistance or sensitivity to corticosteroids. Treatment/dosage: If glucocorticoid anti-inflammatory treatment is used, doses should be slightly increased or an alternative treatment with non-glucocorticoid anti-inflammatory drugs should be chosen.

Treatment efficacy with antiandrogenics

17-α estradiol				
Gene	SNP (transition)	Activating allele	Patient genotype	Pharmacogenetic result
CYP19A1	rs2470152 (C>T)	T	CC	Genetic result: Predisposition to normal CYP19A1 activity. Interpretation: Aromatase (CYP19A1) variants are associated to low conversion of testosterone in estrogens and to high conversion into DHT (hair growth inhibitor). Treatment/dosage: SNP analysis does not indicate a necessity to treat with 17-α Estradiol.

Dutasteride				
Gene	SNP (transition)	Activating allele	Patient genotype	Pharmacogenetic result
SRD5A1	rs39848 (T>C)	C	CT	Genetic result: Predisposition to slightly increased SRD5A1 activity leading to moderately increased DHT levels. Interpretation: Steroid 5α-Reductase 1 (SRD5A1) variants are associated with increased SRD5A1 activity leading to increased DHT levels and hair growth inhibition. Treatment/dosage: Treatment with Dutasteride at normal doses is recommended.

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Finasteride

Gene	SNP (transition)	Activating allele	Patient genotype	Pharmacogenetic result
SRD5A2	rs523349 (C>G)	G	GG	Genetic result: Predisposition to high SRD5A2 activity leading to higher levels of DHT Interpretation: Steroid 5 α -Reductase 2 (SRD5A2) variants are associated with increased SRD5A2 activity leading to increased DHT levels and hair growth inhibition. Treatment/dosage: Treatment with Finasteride at high doses is recommended.

2. Hair care supplements

Vasodilatation and blood circulation

Circulation stimulators

Gene	SNP (transition)	Activating allele	Patient genotype	Pharmacogenetic result
ACE	rs4343 (A>G)	G	AG	Genetic result: Predisposition to an increased Angiotensin conversion activity. Interpretation: Angiotensin-converting enzyme (ACE) variants are associated with increased plasma levels of angiotensin 2, an extremely potent vasoconstrictor. Treatment/dosage: Normal doses of circulation stimulators are recommended, such as Minoxidil, caffeine, Ginkgo biloba, Ginseng or Arginine.

Collagen synthesis

Hair strengthening supplements

Gene	SNP (transition)	Activating allele	Patient genotype	Pharmacogenetic result
COL1A1	rs1800012 (G>T)	T	GG	Genetic result: Predisposition to normal collagen stability. Interpretation: Collagen, type I, alpha 1 (COL1A1) variants are associated with collagen instability. Treatment/dosage: SNP analysis does not indicate the necessity to supplement with hair strengthening composites.

Reduction of IGF-1 levels

Hair strengthening supplements

Gene	SNP (transition)	Activating allele	Patient genotype	Pharmacogenetic result
IGF1R	rs2229765 (G>A)	A	GG	Genetic result: Predisposition to normal IGF-1 levels. Interpretation: Insulin-like growth factor-I (IGF-I) variants are associated with lower plasma IGF-1 levels leading to hair loss. Treatment/dosage: SNP analysis does not indicate the necessity to treat with IGF-1 inducers.

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3. Vitamin, mineral and antioxidant supplements

Vitamins

Vitamin A

Gene	SNP (transition)	Activating allele	Patient genotype	Pharmacogenetic result
CRABP2	rs12724719 (G>A)	A	GG	Genetic result: Predisposition to normal retinoic acid intracellular transport. Interpretation: Cellular retinoic acid-binding protein 2 (CRABP2) variants are associated with lower retinoic acid (vitamin A) intracellular transport. Treatment/dosage: SNP analysis does not indicate the necessity to supplement with vitamin A.

Vitamin B7

Gene	SNP (transition)	Activating allele	Patient genotype	Pharmacogenetic result
BTD	rs13078881 (G>C)	C	GG	Genetic result: Predisposition to normal biotinidase activity. Interpretation: Biotinidase (BTD) variants are associated with low biotin (vitamin B7) uptake from the diet. Treatment/dosage: SNP analysis does not indicate the necessity to supplement with vitamin B.

Vitamin C

Gene	SNP (transition)	Activating allele	Patient genotype	Pharmacogenetic result
SLC23A1	rs33972313 (C>T)	T	CC	Genetic result: Predisposition to higher vitamin C serum level. Interpretation: Solute carrier family 23 member 1 (SLC23A1) variants are associated with lower serum concentration of vitamin C. Treatment/dosage: SNP analysis does not indicate the necessity to supplement with vitamin C. Test for serum levels of vitamin C.

Vitamin B9

Gene	SNP (transition)	Activating allele	Patient genotype	Pharmacogenetic result
MTHFR	rs1801133 (G>A)	A	GG	Genetic result: No predisposition to folate deficiency. Interpretation: Methylene tetrahydrofolate reductase (MTHFR) variants are associated with risk of folate deficiency. Treatment/dosage: SNP analysis does not indicate the necessity to supplement with folate. Test serum levels of folate prior to supplementation.

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Vitamin D

Gene	SNP (transition)	Activating allele	Patient genotype	Pharmacogenetic result
GC	rs2282679 (T>G)	G	TT	Genetic result: Predisposition to normal vitamin D serum levels. Interpretation: Vitamin D-binding protein (GC or DBP) variants are associated with lower vitamin D serum level. Treatment/dosage: SNP analysis does not indicate the necessity to supplement with vitamin D.

Vitamin B12

Gene	SNP (transition)	Activating allele	Patient genotype	Pharmacogenetic result
FUT2	rs602662 (A>G)	G	GG	Genetic result: Predisposition to low vitamin B12 serum level. Interpretation: Galactoside 2-alpha-L-fucosyltransferase 2 (FUT2) variants are associated lower vitamin B12 serum level. Treatment/dosage: Supplementation with vitamin B12 is highly recommended.

Vitamin E

Gene	SNP (transition)	Activating allele	Patient genotype	Pharmacogenetic result
ZPR1	rs964184 (G>C)	C	CC	Genetic result: Predisposition to lower vitamin E serum levels. Interpretation: Zinc Finger Protein ZPR1 variants are associated with low alpha-tocopherol (vitamin E) serum level. Treatment/dosage: Supplementation with vitamin E is highly recommended. Test serum levels of vitamin E prior to supplementation.

Antioxidants

Antioxidants

Gene	SNP (transition)	Activating allele	Patient genotype	Pharmacogenetic result
NQO1	rs1800566 (G>A)	A	AG	Genetic result: Predisposition to lower NQO1 enzyme activity. Interpretation: NAD(P)H dehydrogenase [quinone] 1 (NQO1) variants are associated with lower NQO1 enzyme activity and may have less effective protection against oxidative stress. Treatment/dosage: Supplementation with antioxidants would be recommended. Test serum levels of selenium prior to supplementation.

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Minerals

Magnesium

Gene	SNP (transition)	Activating allele	Patient genotype	Pharmacogenetic result
MUC1	rs4072037 (T>C)	C	CC	Genetic result: Predisposition to lower magnesium serum level. Interpretation: Mucin 1, cell surface associated (MUC1) variants are associated with lower magnesium serum level. Treatment/dosage: Supplementation with magnesium would be highly recommended. Test serum levels of magnesium prior to supplementation.

Zinc sulfate

Gene	SNP (transition)	Activating allele	Patient genotype	Pharmacogenetic result
SLC30A3	rs11126936 (T>G)	G	GG	Genetic result: Predisposition to lower serum zinc level. Interpretation: Solute carrier family 30 member 3 (SLC30A3) variants are associated with lower serum zinc level. Treatment/dosage: Supplementation with Zinc sulfate would be highly recommended. Test serum levels of zinc prior to supplementation.

Iron

Gene	SNP (transition)	Activating allele	Patient genotype	Pharmacogenetic result
TMPRSS6	rs855791 (G>A)	A	TC	Genetic result: Predisposition to slightly reduced serum levels of transferrin and iron. Interpretation: Transmembrane protease, serine 6 (TMPRSS6 or matriptase-2) variants are associated with decreased serum levels of transferrin and iron. Treatment/dosage: Supplementation should be considered. Test serum levels of iron prior to supplementation.

Selenium

Gene	SNP (transition)	Activating allele	Patient genotype	Pharmacogenetic result
DMGDH	rs921943 (T>C)	C	CC	Genetic result: Predisposition to lower selenium serum level. Interpretation: Dimethylglycine dehydrogenase (DMGDH) variants are associated with low selenium serum level. Treatment/dosage: Selenium supplementation should be considered. Test serum levels of selenium prior to supplementation.



V. Methodology

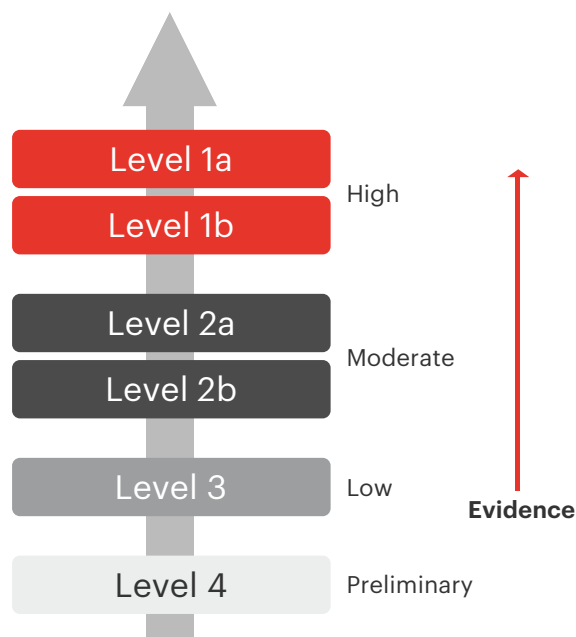
5. Methodology

How were the genetic variants studied selected and evaluated?

The **genetic test** was developed by a multidisciplinary team of medical doctors, pharmacists, geneticists, and programmers, following the highest quality standards. In particular, an expert team specialized in the curation of genetic variants reviewed each variant to ensure that selection, interpretation and impact of variants in the algorithms are based on the highest scientific evidence. Relevant patient's anamnesis (intolerances, diseases, medication, blood pressure, among others) that can affect recommendations was taken into account through medical questionnaires elaborated by health professionals.

- **Level 1A:** Annotation for a variant in medical society-endorsed or implemented in a major health system.
- **Level 1B:** Annotation for a variant where the preponderance of evidence shows an association. The association must be replicated in more than one cohort with significant p-values, and preferably will have a strong effect size.
- **Level 2A:** Annotation for a variant that qualifies for level 2B where the variant is within a Very Important known gene, so functional significance is more likely.
- **Level 2B:** Annotation for a variant with moderate evidence of an association. The association must be replicated but there may be some studies that do not show statistical significance, and/or the effect size may be small.
- **Level 3:** Annotation for a variant based on a single significant (not yet replicated) study or annotation for a variant evaluated in multiple studies but lacking clear evidence of an association.
- **Level 4:** Annotation based on a case report, non-significant study or in vitro, molecular or functional assay evidence only.

Only variants from level 1a to 2b were selected.



How was this test performed?

DNA was extracted from the buccal swab sample provided and was analyzed by our clinical analysis laboratory. DNA was extracted using the KingFisher Flex® robotic extraction system (Thermo Fisher Scientific). The study of the genetic variants was carried out using a custom-designed microfluidic card to measure for the chemiluminescent detection of each of them using TaqMan® technology. TaqMan® technology for genotyping testing is proven and widely used in clinical and research settings. The sensitivity (detection limit) of this study is 99%.

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Genetic test algorithm

The **genetic test** qualitative pharmacogenetic algorithm analyzes single nucleotide polymorphisms (SNPs) associated with metabolic pathways involved in alopecia predisposition and treatment and combines this data with relevant patient history to predict treatment responses and recommends the most appropriate active ingredients.

The **genetic test** is an in vitro diagnostic medical device developed by **Fagron Genomics** and marketed under the CE-IVD mark in conformity with European Directive 98/79/EC and the transitional provisions (article 130) of European Regulation 2017/746.

 **Fagron Genomics S.L.U.**,
SRN: ES-MF-000001092
C/ de les Cosidores, 150
08226 Terrassa, Barcelona (Spain)

What are the limits of this report?

Each genetic marker tested is just one factor that predicts the likelihood of a particular outcome. However, the lifestyle, diet, and environment to which the patient is exposed may impact the expected outcomes. These external factors cannot be taken into account in this report.

The information in this report is not used to diagnose genetic diseases or abnormalities, as it does not predict the risk and likelihood of certain genetic outcomes. It is also not intended to diagnose or cure any disease. The **genetic test** is intended to assist health professionals in making patient-specific care decisions regarding the treatment or prevention of androgenetic alopecia, areata alopecia, and telogen effluvium.

Our clinical laboratory has standard and effective procedures to protect against technical and operational problems. However, problems may occur in the shipment to the laboratory or in the handling of the sample, including, but not limited to, damage to the sample, mislabeling, and loss or delay in receiving the sample. In such cases, the medical laboratory may need to request a new sample.

As with all medical laboratory tests, there is a small chance that the laboratory may provide inaccurate information.

It is the responsibility of the professional who requests a test from us to guarantee the interested party appropriate genetic counseling in accordance with Law 14/2007, of July 3, on Biomedical Research.

Fagron Genomics S.L.U. declines all responsibility derived from the use and interpretation of the results of our tests by the requesting health professional.

Fagron Genomics S.L.U. does not access data identifying the patient from whom the sample comes, so it is also the responsibility of the requesting professional to comply with the applicable data protection regulations.



VI. References

References

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• Sample date: **08-11-2023**

• Date of the results: **13-11-2023**

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